E-Commerce Return Rate Reduction Analysis Using Data Analytics

Introduction:

Product returns are a major issue in e-commerce, causing financial losses and operational challenges. Customers return products due to reasons such as poor quality, incorrect sizing, misleading descriptions, or delivery problems. Analyzing return patterns helps businesses identify problem areas and improve customer satisfaction. This project focuses on studying return behavior across categories, locations, suppliers, and marketing channels.

Abstract:

This project analyzes e-commerce return data to identify factors influencing product returns. Data cleaning and exploratory analysis were performed to compare return rates across different segments. A logistic regression model was used to predict the probability of returns, and high-risk products were identified. An interactive dashboard was created to visualize trends and support decision-making for reducing return rates.

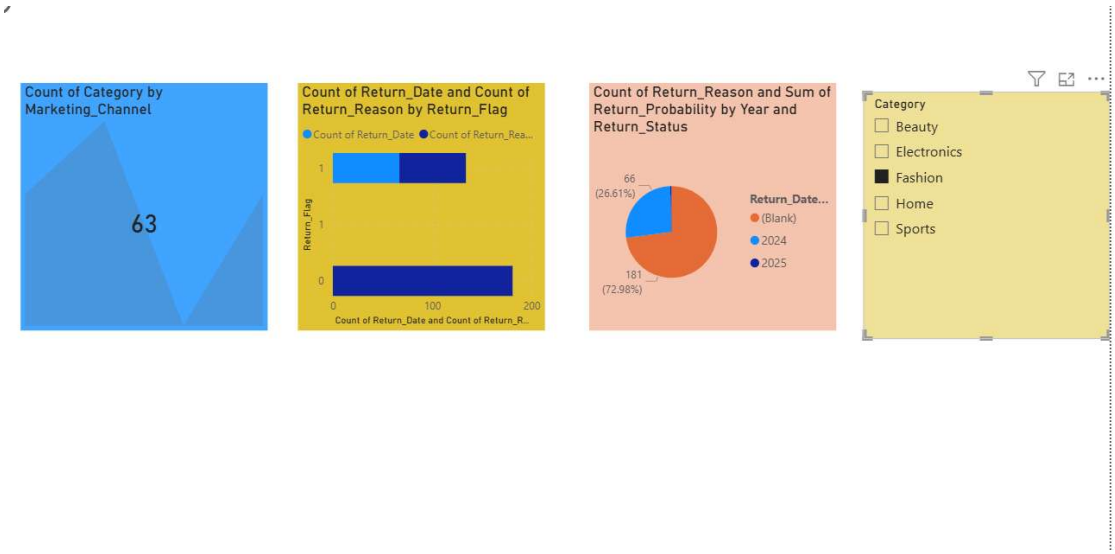
Tools Used:

- Python (Pandas, NumPy) — Data processing
- Scikit-learn — Machine learning model
- Matplotlib / Seaborn — Visualization
- Power BI — Interactive dashboard
- CSV files — Data storage

Steps Involved in Building the Project:

1. Collected order and return datasets
2. Cleaned and preprocessed the data
3. Analyzed return rates by category, location, and channel
4. Identified common return reasons
5. Built a logistic regression model to predict returns
6. Flagged high-risk products
7. Created an interactive Power BI dashboard

Outputs:



Conclusion:

The project shows how data analytics and machine learning can help reduce e-commerce product returns. By identifying patterns and high-risk products, businesses can take preventive actions to improve quality, marketing, and customer experience. The dashboard provides useful insights for better decision-making and operational efficiency.